Phase 5: Scale + Report

Multi-year trend tables, a real report layer -- and testing the whole pipeline against five companies.

What this phase adds

Phases 1-4 work one fiscal year (and, where needed, its prior year) at a time. Phase 5 assembles a whole range of years into pandas DataFrames -- line item by year, ratio by year, articulation check by year -- and renders them as rich terminal tables. pandas enters the project only here, on purpose: it's the one tool actually suited to a line-item x year trend table, and every cell holds a Decimal or None object, never a float, so multi-year tables never quietly round through binary floating point.

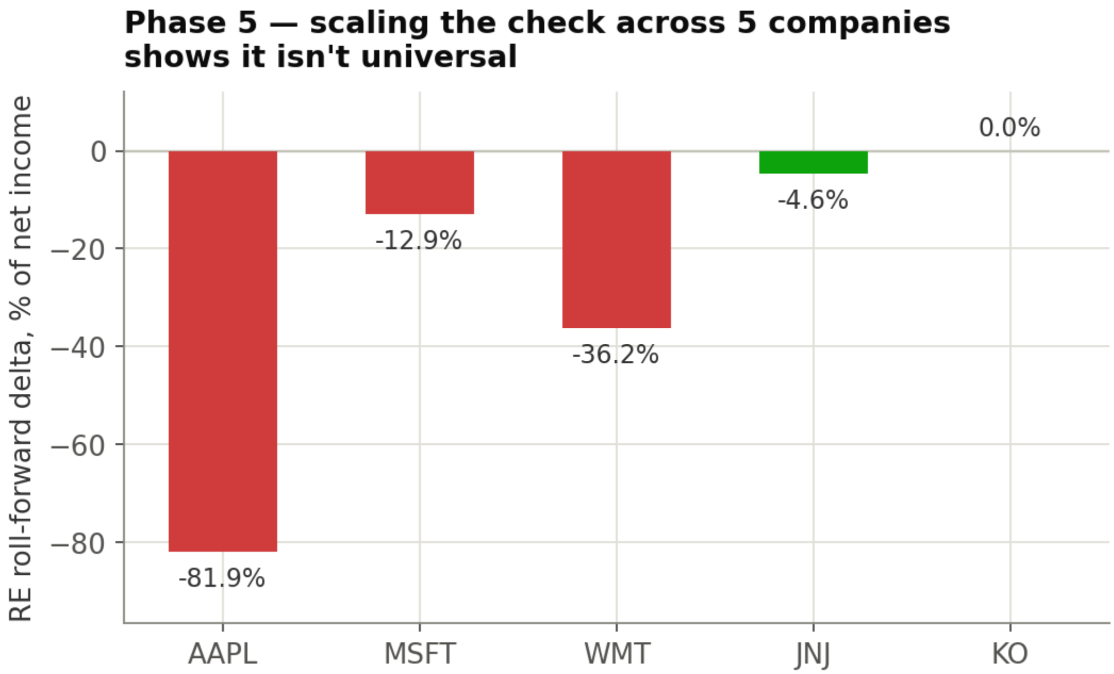
A year that can't be normalized is skipped, not fatal -- the rest of the requested range still renders, with a note about what was skipped. A year at the edge of the requested range with no fetched prior year correctly shows the prior-dependent figures as unmapped rather than reaching outside the range or guessing.

Scale as a debugging tool, not just a feature

The most valuable thing about running five real companies through this pipeline wasn't more data -- it was finding out where the model was quietly wrong. Walmart and Coca-Cola never tag a total "Liabilities" concept at all, which would have silently broken the single most basic check in the whole tool (Assets = Liabilities + Equity) for two of five companies tested. That kind of gap doesn't show up testing one company carefully -- it shows up testing several companies casually. After adding a derivation (total_liabilities = total_liabilities_and_equity - total_equity when the direct tag is absent), all five companies tie exactly.

Scale also answers a question one company can't: is this pattern universal?

Phase 3's retained-earnings finding on Apple -- a growing delta from buyback-driven treasury stock retirement -- could have been an Apple-specific quirk. Running the same check across five companies' latest fiscal years, scaled to each company's own net income for a fair comparison, shows it isn't universal at all: Johnson & Johnson and Coca-Cola tie almost exactly, while Apple, Microsoft, and Walmart show real, material drift.



Retained-earnings roll-forward delta, latest fiscal year per company, as a percentage of that year's net income.

Company	Delta ($B)	Delta (% of net income)
Apple	-91.7	-81.9%
Microsoft	-13.2	-12.9%
Walmart	-7.9	-36.2%
Johnson & Johnson	-1.2	-4.6%
Coca-Cola	0.0	0.0%

KEY TAKEAWAY

One company's result can look like a bug, an outlier, or a lesson -- you often can't tell which from a single data point. Five companies' results turn the same check into a real comparison, and that's exactly when a pattern like "buybacks distort the RE roll-forward" stops being a guess and becomes something you've actually shown.